- **DataFrame:** A two-dimensional, tabular data structure that is similar to a spreadsheet or SQL table. It consists of rows and columns, with the ability to hold different data types in each column.

These data structures enable intuitive handling of structured data, making Pandas an indispensable tool for data scientists.

Loading Data with Pandas

Pandas offers a variety of methods for loading data into a DataFrame. The most commonly used is the read_csv() function, which enables seamless reading of CSV files.

With just a single line of code, you can import your dataset into Python: import pandas as pd

data = pd.read_csv('dataset.csv')
print(data.head())

The head() method is particularly useful, as it provides a quick glimpse of the first few rows of your dataset, helping you understand its structure and contents.

Beyond CSV Files

While CSV is a popular format, Pandas supports a wide range of file types, making it highly versatile for working with diverse datasets. Some examples include:

- **Excel Files:**

data = pd.read_excel('dataset.xlsx')

- **JSON Files:**

data = pd.read_json('dataset.json')

- **HTML Tables:**

data = pd.read_html('http://example.com/table.html')[0]

- **SQL Databases:**

import sqlite3

conn = sqlite3.connect('database.db')
data = pd.read_sql('SELECT * FROM table_name', conn) This flexibility ensures that you can load data from virtually any source, simplifying the data acquisition process.